THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM ONE ANNUAL EXAMINATION
032 CHEMISTRY

Time: 2½ Hours **November 2025**

Instructions

1. This paper consists of sections **A**, **B** and **C** with a total of ten questions.
2. Answer **ALL** questions in the spaces provided.
3. All writing must be in blue or black ink, except drawing, which must be in pencil.
4. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
5. Write your **Examination Number** at the top right corner of every page.

QUESTION #	MARKS	INITIALS OF EXAMINER
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
Total		

SECTION A (15 MARKS)

1. For each of the items (i)–(x), choose the correct answer from the given alternatives and write its letter beside the item number in your answer booklet.

(i) Which of the following is a physical change?
A. Burning wood.
B. Rusting of iron.
C. Melting ice.
D. Cooking food.

(ii) The following are important laboratory safety measures to obey, except:
A. Properly labeling all chemicals
B. Storing food in a refrigerator or freezer where chemicals are stored
C. Labeling and locking all storage areas.
D. Making sure fire extinguishers are in good condition.

(iii) The instrument used to measure the volume of a liquid is a:
A. Thermometer.
B. Measuring cylinder.
C. Tripod stand.
D. Balance.

(iv) The following are examples of substances that change directly from solid to gaseous state when heated:
A. Sodium chloride and sugar
B. Ammonium chloride and iodine
C. Calcium carbonate powder and ink
D. Sodium carbonate crystals and copper sulphate powder

(v) Which of the following is a mixture?
A. Distilled water.
B. Pure gold.
C. Air.
D. Sodium chloride.

(vi) The process of separating an insoluble solid from a liquid using filter paper is called:
A. Distillation.
B. Filtration.
C. Evaporation.
D. Sublimation.

(vii) The main difference between a compound and a mixture is that a compound:
A. Can be separated by physical means.
B. Has a definite chemical composition.
C. Has its components randomly arranged.

D. Shows the properties of its components.

(viii) The Liebig condenser is an apparatus used for:
A. Collecting gas
B. Delivering gas during collections
C. Separating mixtures of two liquids
D. Cooling hot vapours

(ix) Which branch of Chemistry deals with the study of carbon and its compounds?
A. Inorganic chemistry
B. Organic chemistry
C. Biochemistry
D. Analytical chemistry

(x) The type of fire extinguisher used to put out fires caused by flammable liquids is a:
A. Water extinguisher.
B. Carbon dioxide extinguisher.
C. Foam extinguisher.
D. Dry powder extinguisher.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)

2. Match the items in **List A** with their corresponding descriptions in **List B**.

LIST A	LIST B
(i) Bunsen burner	A. Used for holding chemicals during heating.
(ii) Test tube holder	B. A source of heat in the laboratory.
(iii)Beaker	C. Used for heating small amounts of liquid.
(iv)Funnel	D. Used for holding a test tube during heating.
(v) Conical flask	E. Used for separating an insoluble solid from a liquid
	F. Used for holding and swirling chemicals.

(i)	(ii)	(iii)	(iv)	(v)

SECTION B (70 MARKS)

3. (a) Explain the difference between an element and a compound.

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(b) Give two examples of common laboratory hazards and how to prevent them.

(i) ……

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(ii) ……

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4. (a) You have been appointed in a village meeting as an expert of Chemistry to explain to the villagers four applications of Chemistry in our daily life on the following fields:

(i) Construction material industry

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(ii) Agriculture

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(iii) Medicine and pharmacy

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(iv) Transportation

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(b) A form one student at Seypalm secondary school was asked to mention four professional careers he/she may attain through studying Chemistry. What correct answers can we get from him?

(i) ……

(ii) ……

(iii)……

(iv)…….....

5. (a) Explain the term chromatography and state its principle.

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(b) Suggest a suitable method for separating the following mixtures:

(i) Sand and salt.

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(ii) Water and kerosene.

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(iii)Alcohol and water.

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6. (a) Define the following terms:

(i) Periodicity

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(ii) Periodic table

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(b) Give a brief description of the following groups of elements by stating the group in which they are found, their main features, and two examples of representative elements for each of the groups.

(i) Alkaline metals

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(ii) Alkaline earth metals

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(iii)Noble gases

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7. (a) Before starting to fight the fire, it is important to know the size and type of the fire that you are going to put off. In this case, so as to effectively fight fires, fires are classified based on the burning materials. Describe class B and class C fires and the kind of firefighting materials that are used to effectively extinguish those two types of fires respectively.

(i) Class B fire:

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(ii) Class C fire:

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(b) Explain the three components that together create the necessary condition for a fire to continue or start to burn if these components are present:

(i) ……

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(ii) ……

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(iii)……

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8. Laboratory rules are established in order to avoid accidents, injuries and damages to property. State any ten important laboratory rules:

(i) ……

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(ii) ……

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(iii)……

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(iv)……

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(v) ……

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(vi)……

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(vii)……

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(viii) ……

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(ix)……

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9. (a) Define ‘solution’ and give two examples of solution

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(b) Give four differences a mixture and a compound

Mixture	Compound
(i)	
(ii)	
(iii)	
(iv)	

SECTION C (15 MARKS)

10. Most students know that the changes in the states of matter is a subtopic topic taught in Chemistry, however, they don’t know how the knowledge of the changes in the states of matter is useful in our daily life. As a student who studies Chemistry, giving six reasons, educate your fellow students about the importance of changing one state of matter to another.

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